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Q1. Arrays, Array Applications and Structures

CODE

```
#include <stdio.h>
int main()
{
    char str[100];
    int i, j, count;
    scanf("%s", str);
    for(i = 0; str[i] != '\0'; i++)
    {
        count = 0;
        for(j = 0; str[j] != '\0'; j++)
        {
            if(str[i] == str[j])
                count++;
        }
        if(count == 1)
        {
            printf("first non-repeated character:%c", str[i]);
            return 0;
        }
    }
    printf("no non-repeated character found");
    return 0;
}
```

INPUT

swiss

OUTPUT

(no output)

Q2. Arrays, Array Applications and Structures

CODE

```
#include<stdio.h>
int main(){
    int n,a[100],key;
    int low,high,mid;
    scanf("%d",&n);
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    scanf("%d",&key);
    low=0;
    high=n-1;
    while(low<high){
        mid=(low+high)/2;
        if(a[mid]==key){
            printf("%d",mid);
            return 0;
        }
        if(a[low]<=a[mid]){
            if(key>=a[low]&&key<a[mid])
                high=mid-1;
            else
                low=mid-1;
        }
    }
    printf("-1");
    return 0;
}
```

INPUT

```
7
1 2 3 4 5 6 7
0
```

OUTPUT

(no output)

CODE

```
#include <stdio.h>

int main() {
    int n, a[100][2], i, j, t, s, e;

    scanf("%d", &n);

    for(i = 0; i < n; i++) {
        scanf("%d %d", &a[i][0], &a[i][1]);
    }

    // Sort intervals by start time
    for(i = 0; i < n - 1; i++) {
        for(j = i + 1; j < n; j++) {
            if(a[i][0] > a[j][0]) {
                t = a[i][0];
                a[i][0] = a[j][0];
                a[j][0] = t;

                t = a[i][1];
                a[i][1] = a[j][1];
                a[j][1] = t;
            }
        }
    }

    s = a[0][0];
    e = a[0][1];

    for(i = 1; i < n; i++) {
        if(a[i][0] <= e) {
            if(a[i][1] > e)
                e = a[i][1];
        } else {
            printf("%d %d\n", s, e);
            s = a[i][0];
            e = a[i][1];
        }
    }

    printf("%d %d", s, e);

    return 0;
}
```

INPUT

```
4
1 3
2 6
8 10
15 18
```

OUTPUT

(no output)

CODE

```
#include<stdio.h>
#include<string.h>
int main() {
    char str[100], temp;
    int freq[100];
    int i, j, len;
    scanf("%s", str);
    len = strlen(str);
    for(i = 0; i < len; i++) {
        freq[i] = 1;
        if(str[i] == '0')
            continue;
        for(j = i + 1; j < len; j++) {
            if(str[i] == str[j]) {
                freq[i]++;
                str[j] = '0';
            }
        }
    }

    for(i = 0; i < len - 1; i++) {
        for(j = i + 1; j < len; j++) {
            if(freq[i] < freq[j]) {
                int t = freq[i];
                freq[i] = freq[j];
                freq[j] = t;

                temp = str[i];
                str[i] = str[j];
                str[j] = temp;
            }
        }
    }
    for(i = 0; i < len; i++) {
        if(str[i] != '0') {
            for(j = 0; j < freq[i]; j++)
                printf("%c", str[i]);
        }
    }

    return 0;
}
```

INPUT

Tree

OUTPUT

(no output)